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January 12, 1976
SUPERSEDING
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FEDERAL SPECIFICATION

TAPE, PAPER, COMPUTING MACHINE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers four types of paper tapes that are used with various adding and computing machines, including electronic printing calculators.

1.2 Classification.

1.2.1 Types. Paper tapes, in accordance with this specification, shall be furnished in the following types (see 6.2):

Type I - Chemical Wood.
Type II - Fibrous Cellulosic Material.
Type III - Chemical Transfer, Carbonless.
Type IV - Chemical Self-Imaging, Carbonless.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

DDD-R-271 - Ribbons; Computing and Recording Machines.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks, and the Index of Federal Specifications and Standards, from established distribution points in their agencies.)

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Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 401 - Fiber Analysis of Paper and Paperboard.
T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets, and Related Products.
T 403 - Bursting Strength of Paper.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 411 - Thickness (Caliper) of Paper and Paperboard.
T 414 - Internal Tearing Resistance of Paper.
T 452 - Brightness of Paper and Paperboard.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Tariff Order Section, 1616 P Street, NW., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Materials used shall conform to the requirements specified herein. A certificate shall be furnished by the contractor to the procurement officer stating that the computing machine tapes shall contain no poly-chlorinated biphenyls (PCB's).

3.1.1 Type I. The paper stock shall be 100 percent chemical wood pulp, free from any ground wood pulp.

3.1.2 Type II. The paper stock shall be made from fibrous cellulosic material which may include reclaimed fibers.

3.1.3 Type III. Carbonless paper tape, type III, shall be made from fibrous cellulosic material and shall be of the chemical transfer type to be used with machine ribbon.

3.1.4 Type IV. Carbonless paper tape, type IV, shall be made from fibrous cellulosic material and shall be of the chemical transfer type to be used without machine ribbon.

3.2 Physical properties. The physical properties of paper tapes shall be as specified in table I when tested in accordance with 4.3. The paper tape shall be white.

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TABLE I. Physical Requirements

Property	Type I	Type II	Types III and IV
Basis Weight, pounds (17 x 22 - 500) ± 5%	16	16	12-16*
Bursting Strength, pounds per square inch (psi) (min.)	16	12	12
Tearing Strength, grams (min.)			
Cross Direction	50	28	40
Machine Direction	44	23	30
Brightness, percent (%)(min.)	75	65	72
Thickness, inches	.0029-.0039	0.0035-0.0045	.0025-.0040
Stiffness (mg.)(min.)			
Cross Direction	75	75	45
Machine Direction	35	35	25

*The ± 5% tolerance does not apply.

3.3 Widths. The width of the paper tape shall be 2, 2-1/4, 2-1/2, 2-3/4, 3, 3-1/16, or 3-1/8 inches with a tolerance of plus or minus 1/32 inch, as specified (see 6.2).

3.4 Rolls. The paper tape shall be wound with the felt side out on a core evenly and tightly to produce a roll of an overall outside diameter of 3-1/2 inches maximum, and containing not less than 200 feet. The paper tape shall be one continuous length without splices, joints, seams, or tears.

3.4.1 Cores. The core may be wooden, fiber, hard wound paper, plastic, or any other materials which are not susceptible to cracking, splitting, or shrinkage under ordinary conditions of shipment and use. The round core shall be approximately 7/8 inch in outside diameter with a 3/8 to 7/16 inch diameter spindle hole in the center of the core. The length of the core shall be the same as the width of the tape, with a ± 1/32 inch tolerance.

3.4.2 Construction. The inside end of the paper shall be tucked or attached to the core at the time of winding. The outside end of the paper tape shall be sealed in such a manner as to allow enough unglued paper to serve as a pull tab for unsealing the roll at time of initial use, or spot-sealed by use of button style, latex-coated, or gummed labels.

3.4.3 Warning signal. A warning signal, 3 to 10 feet long, not less than 1/8 inch wide, shall be applied starting within 5 feet from the core end of the tape to warn the operator of the approaching end of the tape. The color of the warning signal shall be a distinguished color which will not affect the readability of the printed figures.

3.5 Printing quality. When printed by means of a computing machine ribbon, the ink shall be quickly absorbed without smearing or blurring of the printed figures, when tested in 4.3.1.3.2, for types I, II, and III. Image production shall be by chemical interaction on the carbonless types III and IV tapes and the prints shall be legible.

3.6 Writing quality. The paper tape shall be suitable for writing on with ink, when tested in 4.3.1.3.1.

3.7 Workmanship. The paper tape shall be free from any defects which may affect its serviceability and shall be clean and free from fluff, lint, paper dust, grit, spots, slime spots, calendered scorches, wrinkles, folds, holes, tears, and other imperfections. The finish of the paper tape shall be smooth, noncockled, and nonglazed. The rolls of paper tape shall be smoothly cut without nicks or ragged edges.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated. A lot shall consist of all rolls of paper tape of one size and type offered for delivery at one time and a unit of production shall be one roll unless otherwise specified.

4.1.2 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications and standards.

4.1.2.1 Component testing. Each lot of component material shall be tested for the characteristics specified in section 3 in accordance with the test methods specified in 4.2. The lot shall consist of all materials of the same type, manufactured by the same source, under essentially the same conditions, and identified as used in the production of the quality of end item rolls submitted for inspection at one time.

4.1.2.1.1 Evidence of compliance. In lieu of utilizing his own facilities, the contractor may submit a report of analysis of the component material from a laboratory acceptable to the Government as evidence of compliance with the specification requirements. The report of analysis shall include, in addition to the actual test data, the manufacturer's name, name and address of the testing laboratory, date of test, and quantity of material represented by the test report. When certificates of compliance or reports of analysis are submitted, the Government reserves the right to check test such items to determine the validity of the certificate or analysis.

4.2 Testing of components. The paper shall be tested for the applicable characteristics specified in table II. The test shall be made in accordance with the indicated TAPPI methods. The lot size for the purpose of determining the sample size shall be expressed in pounds of paper, used in the manufacture of the paper tape. The sample unit shall consist of not less than 1,000 square inches of base paper. The sample size shall be drawn in accordance with level S-1 of MIL-STD-105.

Test results shall include all values upon which the results are based, and there shall be no failures to meet the composite or lot average requirements. The composite sample shall consist of approximately equal amounts drawn from all the sample units combined into a single sample. Except where otherwise specified, all samples shall be conditioned and tested under uniform atmospheric conditions, as specified in TAPPI Method T 402.

TABLE II. Instructions for Testing Components

Characteristics	Specification Requirement	Reference Test Method	Requirement Applied to	Results Reported to Nearest
Paper stock	3.1	T 401	Composite	1 percent
Atmospheric conditions	4.2	T 402	All testing	
Basis weight	3.1	T 410	Lot average	0.1 pound
Bursting strength	3.1	T 403	Lot average	1 psi
Tearing resistance	3.1	T 414	Lot average	1 gram
Thickness	3.1	T 411	Lot average	0.0001 inch
Brightness	3.2	T 452	Lot average	1 percent
Stiffness	4.2.1	-	Lot average	0.1 mg.

4.2.1 Stiffness. The stiffness is to be determined on a Gurley Stiffness Tester using manufacturer's instructions. Size of specimen shall be 1" x 3-1/2", testing 10 specimens in both machine and cross directions.

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4.3 End item inspection.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQL's) set forth therein.

4.3.1.1 Examination of preparation for delivery. An examination shall be made to determine whether the packing and marking comply with the requirements of section 5. Defects shall be scored as specified below. Sampling shall be in accordance with MIL-STD-105. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper, or missing.

4.3.1.1.1 Examination of closure, waterproofing, and banding. When shipping containers are required to comply with PPP-B-636, examination of closure, waterproofing, and banding shall be in accordance with the appendix of that specification.

4.3.1.2 Examination of end item. Tapes shall be examined for defects as listed below. The lot size shall be expressed in units of rolls. The inspection level shall be S-4 and the AQL shall be 4.0 expressed as percent defective. The sample unit shall be one roll of paper tape. Rolls shall be obtained from containers selected in 4.3.1.1.

Examination	Defect
Dimension	Width not within tolerance, diameter of roll too large, core center hole diameter not as specified. Roll contains less than 200 feet of paper. Core length not as specified.
Roll	Paper tape not one continuous length. Has splices, joints, or seams; warning signal missing; tape not wound felt or smooth side out; core not rigid; core cracked, splintered, or shrunk; tape not tucked or attached to core; outside end of tape not sealed properly; not wound evenly or tightly.
Color	Not white.
Workmanship	Not clean or free from fluff, lint, paper dust, grit, dirt, spots, slime spots, calender scorches, fold, holes, tears, or other imperfections; edges not smooth, have nicks or ragged cut.

4.3.1.3 Testing of end item. Rolls of paper tape shall be selected from containers selected in 4.3.1.1 from each lot in accordance with inspection level S-2. The lot size shall be expressed in units of rolls. Acceptance or rejection shall be based upon lot average of the selected samples. If any roll fails the test in 4.3.1.3.1 and 4.3.1.3.2, the lot shall be rejected.

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4.3.1.3.1 Writing quality. Write or draw several lines of words or characters on both sides of the paper using a pen with a firm medium point and a 1 percent solution of either C.I. Acid Red 1 (Color Index Number 18050) or C.I. Acid Green 3 (Color Index Number 42085)(see 6.3). To be considered satisfactory, the lines or characters shall be clear-cut and legible without feathering or spreading.

4.3.1.3.2 Printing quality. Print numbers on the sample rolls of paper tape using an adding machine with a ribbon conforming to the requirements of DDD-R-271. The paper tape shall be examined for ink smearing or blurring.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Shipping container. Paper tape in roll form, of one description only, in quantities as specified in table III, shall be packed in a box conforming to PPP-B-636, class weather-resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix to PPP-B-636.

TABLE III. Quantities per Shipping Container

Width	Quantity
Under 2-1/4 inches	100
2-1/4 inches and over	50

5.1.2 Level B. Shipping container. Paper tape shall be packed as specified in 5.1.1 except that the box shall be class domestic.

5.1.3 Level C. Shipping container. Paper tape in quantities as specified (see 6.2), shall be packed in a container complying with the Uniform or National Motor Freight Classification, as applicable. The containers shall be packed to assure carrier acceptance and safe arrival at destination.

5.2 Marking.

5.2.1 Civil agencies. Shipping containers shall be marked in accordance with Fed. Std. No. 123 (see 6.2).

5.2.2 Military agencies. In addition to any special marking required by the contract or order, interior boxes, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use. The rolls of paper tape are intended for general computing machine work. The general name for this item is "adding machine tape" or "adding machine paper rolls".

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of roll (see 1.2.1).
(c) Width (see 3.3).
(d) Applicable level of packing (see 5.1).
(e) Marking requirements (see 5.2).

6.3 Color designations are those listed in the Technical Manual of the American Association of Textile Chemists and Colorists. Acid Red 1, Color Index Number 18050, may be obtained from Textile Aniline and Chemical Company. Acid Green 3, Color Index Number 42085, may be obtained from the following: Sandoz, Incorporated; National Aniline Division of Allied Chemical Corporation; Geigy Chemical Corporated; and American Cyanamid Corporation.

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MILITARY INTERESTS:

User Activities:

Army - GL
DOD coordination has been waived.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

PREPARING ACTIVITY: GSA-FSS

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